

Magnitude of $\mathcal{C}$: The Effective Number of Roles, Redundancy-Corrected

A central mathematical invariant unique to enriched categories is their **magnitude**—a numerical quantity capturing the “effective size” of the category by discounting distributional overlap between objects.

For an enriched category with n objects, let Z be the $n \times n$ similarity matrix where $Z_{ij} = \mathcal{C}(i, j)$. The categorical magnitude is:

$$|\mathcal{C}| = \sum_{i,j} (Z^{-1})_{ij} \tag{1}$$

assuming Z is invertible. This magnitude metric connects to information theory:

- ▶ For a *discrete* category (no non-trivial relationships), $|\mathcal{C}| = n$ (the number of objects)
- ▶ For a highly connected category, $|\mathcal{C}| < n$ (objects are